

MIE245 - Winter 2026

Lab 8 - Graph Exploration

Deadline: 17 March 11:59 pm

Submission Instructions

- **Reminder:** You cannot claim work that was completed by others or generative AI as your own. Submit a completed **readme** template file outlining what code you contributed and what code was completed by other students, people, tools, and/or leveraged from websites.
- Complete the starter code file `MIE245_Lab_8_Template.py` provided to you.
- Submit your completed Python file with the naming format `MIE245_Lab_8_Template_<utorid>.py` on Quercus, with utorid replaced by your UTORid in all lower-case characters.
- Ensure you use your UTORid, i.e., what you sign into Quercus and not other identifiers such as your student number.
- Compress your completed template files to a zip file and upload the zip file to Quercus.

Lab Instructions

A satellite snapshot of a forest can be represented as a two-dimensional (2D) grid. Each cell is labeled as either 1 if a tree in that cell is burning, or 0 otherwise. Two burning cells belong to the same **fire zone** if they are adjacent **horizontally or vertically** (up, down, left, right). Diagonal adjacency does *not* connect zones. A few example grids and their corresponding count of fire zones are presented in Grid 1, 2 and 3.

0	1
1	0

Grid 1: 2x2 grid containing 2 fire zones.

Write a function `count_fire_zones(grid)` that takes as input an $M \times N$ **grid** of 0s and 1s, where $M, N \in \mathbb{Z}_+$, represented as a 2D Python list. The function should return the number of distinct fire zones in the grid. Your algorithm should process each cell a constant number of times, resulting in an overall time complexity of $O(MN)$. You may modify the input **grid** during processing or use an auxiliary data structure.

1	1	0	0
1	0	0	1
0	0	1	1
0	0	0	0

Grid 2: 4x4 grid containing 2 fire zones.

1	1	0	0	1
1	0	0	1	0
0	0	1	1	1
1	1	0	0	1

Grid 3: 4x5 grid containing 4 fire zones.

Auto-Grader Instructions

You can also access an example of the auto-grading script that will be used to test your submission (`MIE245_Lab_8_Autograder.py`). Please follow the steps below to run the auto-grader for the provided test case:

1. Place the auto-grader script in a new folder (for example: `lab8`).
2. Inside the newly created folder, create a folder called `submissions`.
3. Place your script (`MIE245_Lab_8_Template_<utorid>.py`) in the `submissions` folder.
4. Run the auto-grader script and open the `grades.csv` file to check if you passed the test cases.

Note that this script contains simple test cases, so you may not have a completely correct solution even if you pass these test cases.